OPEN-SOURCE AUDIT REPORT

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Software Chosen: Git

INTRODUCTION

In the current technological landscape, modern computing has evolved far beyond a mere collection of isolated applications or closed-off proprietary silos. Instead, it is built upon deep, interconnected layers of global and transparent collaboration that span across continents. We must also recognize that the vast majority of the tools we depend upon today, such as the massive cloud-based servers that power global commerce and the complex compilers that translate the logic of the human brain into the language of the machine itself, were not developed in corporate isolation. Rather, they are all part of the vast and collective open-source software ecosystem that belongs to the community at large. The open-source software itself plays a central and absolutely indispensable role in this modern ecosystem. It is the fundamental engine that allows developers to build, modify, and distribute these critical tools for free and without the burden of the heavy legal and financial restrictions that normally come with such software. Among the myriads of fundamental tools that make up this ecosystem, Git is a sophisticated and revolutionary Distributed Version Control System that has fundamentally and utterly redefined the way in which the entire world tracks changes.

Git has enabled the world to collaborate across borders and time zones in such a way that maintains the integrity and security of massive and complex projects that involve thousands of contributors. In our current era, nearly every professional software development lifecycle is anchored by Git, earning it the reputation of being the "silent foundation" of the global software industry. The purpose of this comprehensive report is to look past the basic, surface-level functions of Git and instead explore it as a living, breathing representation of open-source philosophy. Throughout this audit, we will examine Git's dramatic historical origin, which was born out of a period of intense crisis for the Linux kernel community. We will be analyzing its rigorous

licensing under the GPL v2, a legal framework specifically designed to protect the "digital commons" from privatization. Furthermore, this report delves into the ethical implications of community driven labor, Git's specific technical footprint within the Linux operating system, and the massive ripple effect it has created across the global FOSS ecosystem. To provide a complete picture, practical knowledge is demonstrated through five custom-built shell scripts. These scripts reflect real world automation and administrative tasks, proving that the true philosophy of open source is best understood through direct, hands-on application and experimentation. By auditing Git, we gain a profound and necessary insight into the very nature of software freedom and the collaborative spirit that continues to power our digital world.

PART A

ORIGIN AND PHILOSOPHY

A1.

THE PROBLEM GIT WAS CREATED TO SOLVE

Git was created in 2005 during a pivotal and high stakes turning point in the history of Linux development. Before Git, the Linux kernel community relied on a proprietary version control system called BitKeeper. BitKeeper provided its services for free to the kernel developers under specific conditions, it was fundamentally not open source. This created a dangerous technical dependency and growing philosophical tension within a community dedicated to software freedom. When licensing disputes eventually emerged between the BitKeeper's vendor and the community, access to the tool was revoked. This made Linux kernel project one of the largest and most complex codebases in the world without a proper system to manage its contributions. In response Linus Torvalds decided to solve the problem permanently by building a new tool from scratch. His requirements were strictly designed to prevent such a crisis from happening again:

•High Speed: Operations had to be near instantaneous regardless of project size.

•Distributed Structure: Every developer must have a full copy of the repository to eliminate central failure points.

•Strong Data Integrity: Ensuring that the history of the code could not be corrupted or altered.

•Massive Scale: Ability to handle thousands of parallel branches and contributors efficiently.

Git's introduction of the distributed model eliminated dependency on any single central server drastically improving both reliability and speed of development. It shifted the paradigm of collaboration from centralized control to a model of distributed contribution and making software development more flexible and democratic.

A2.

THE LICENSE:

Git is legally protected and shared under the GNU General Public License version 2(GPLv2). This license is the true cornerstone of its success as it legally enforces the four fundamental freedoms that define free software:

Freedom to Run: User has the absolute right to execute the program for any purpose.

Freedom to Study: User can inspect source code to understand its inner logic.

Freedom to Modify: Users are free to change code so it suits your own specific needs.

Freedom to Distribute: User has the right to share copies of the software and any modifications you have made with the world.

Git fully satisfies every one of these freedoms. Critically, the GPL is what we call a "copyleft" license, which ensures that if you distribute a modified version of the software, it must also remain open source. This legal safeguard is vital preventing companies from taking the community's collective work, tweaking it for their own gain, and then locking it away as closed-source proprietary software. While other licenses, like MIT, offer more flexibility for proprietary reuse, the GPL prioritizes a strict, unyielding enforcement of openness. For Git, the phrase "free as in freedom" means that users are never trapped by vendor ownership ; instead, they maintain total control over the tools they use every day.

A3.

ETHICS OF OPEN SOURCE

Open source represents more than just a specific development model. It is a belief system which has value for transparency, global collaboration and universal accessibility.

•Arguments for Open Source: It encourages rapid innovation through shared knowledge, builds deep technical trust through code visibility and reduces the risk of corporate monopolies over essential tools.

•Arguments against Open Source: It presents revenue challenges for creators. Carries a risk of misuse by bad actors and places a maintenance burden on community volunteers.

Ethical participation in FOSS ecosystem requires a balance between profit and contribution. Companies that build on open-source foundations have a responsibility to "give back" by contributing code, funding, or documentation to community. Ultimately, open source enables developers to build on existing shoulders of giants accelerating human progress without any need to reinvent the wheel for every new project.

PART B

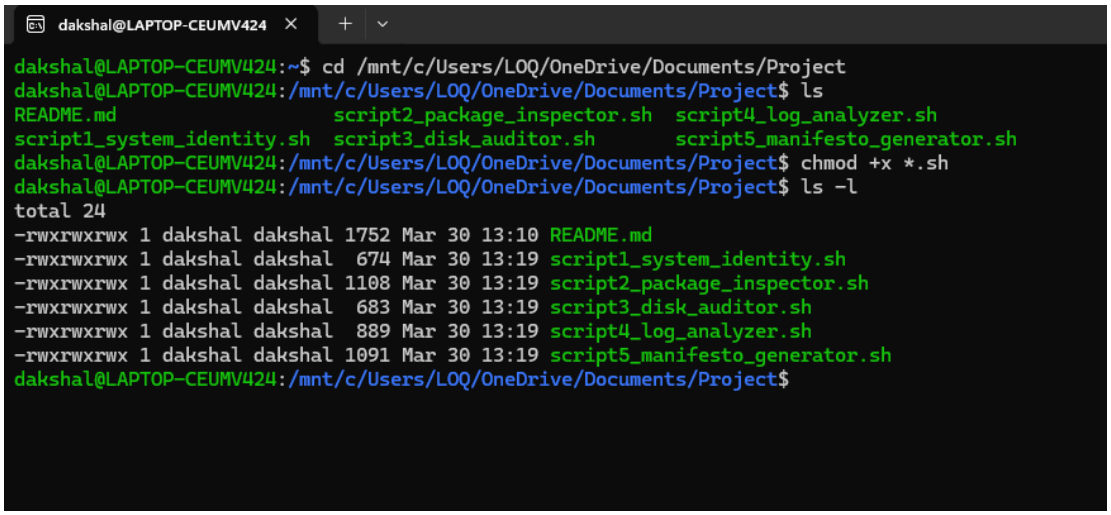
LINUX FOOTPRINT

As a tool created by the founder of Linux for the Linux kernel, Git integrates with exceptional natural efficiency into Linux systems. Its technical footprint is defined by its transparency and its adherence to the Linux philosophy of "small tools doing one thing well."

Installation & Management:

- **Package Managers:** Git is accessible via standard commands like apt install git for Ubuntu systems or yum install git for RHEL/CentOS systems.
- **Key Directories:**
    - /usr/bin/git: The location of the main binary.
    - /etc/gitconfig: System wide configuration settings.
    - ~/.gitconfig: User specific preferences and identities.

Git is not a background service or a "daemon" which consumes resources when idle. Instead, it is an on-demand utility that executes only when a user calls commands such as clone, commit or push. It runs strictly under the permissions of current user ensuring security and permission controls are maintained by the underlying Linux OS. Regular updates and security patches are pushed through the system's package manager by keeping the tool secure in a professional environment.



```
dakshal@LAPTOP-CEUMV424
dakshal@LAPTOP-CEUMV424:~$ cd /mnt/c/Users/LOQ/OneDrive/Documents/Project
dakshal@LAPTOP-CEUMV424:/mnt/c/Users/LOQ/OneDrive/Documents/Project$ ls
README.md                   script2_package_inspector.sh  script4_log_analyzer.sh
script1_system_identity.sh  script3_disk_auditor.sh       script5_manifesto_generator.sh
dakshal@LAPTOP-CEUMV424:/mnt/c/Users/LOQ/OneDrive/Documents/Project$ chmod +x *.sh
dakshal@LAPTOP-CEUMV424:/mnt/c/Users/LOQ/OneDrive/Documents/Project$ ls -l
total 24
-rwxrwxrwx 1 dakshal dakshal 1752 Mar 30 13:10 README.md
-rwxrwxrwx 1 dakshal dakshal  674 Mar 30 13:19 script1_system_identity.sh
-rwxrwxrwx 1 dakshal dakshal 1108 Mar 30 13:19 script2_package_inspector.sh
-rwxrwxrwx 1 dakshal dakshal  683 Mar 30 13:19 script3_disk_auditor.sh
-rwxrwxrwx 1 dakshal dakshal  889 Mar 30 13:19 script4_log_analyzer.sh
-rwxrwxrwx 1 dakshal dakshal 1091 Mar 30 13:19 script5_manifesto_generator.sh
dakshal@LAPTOP-CEUMV424:/mnt/c/Users/LOQ/OneDrive/Documents/Project$
```

Figure 1: Verification of project files and granting of executable permissions using chmod +x in the Ubuntu WSL environment.

PART C

FOSS ECOSYSTEM

Git is core of a massive and thriving ecosystem. Its reliability stems from its reliance on fundamental Linux shell utilities and standard system libraries. By providing a stable and distributed way to manage code, Git directly enabled the creation of industry defining platforms like **GitHub** and **GitLab**.

These platforms made modern **DevOps** practices such as Continuous Integration and Continuous Deployment (CI/CD), which rely on Git to automate the testing and delivery of software. The governance of Git itself is an open model driven by global contributors through mailing lists and a meritocratic structure. Git is the undisputed center of collaborative software development worldwide.

PART D

OPEN-SOURCE VS PROPRIETARY

When comparing Git to proprietary alternatives like the original BitKeeper or Perforce, the advantages of open-source model become clear:

- **Cost:** Git is free to use whereas proprietary systems often require very expensive per user licenses.
- **Security:** Git is transparent as any developer can audit the code for vulnerabilities. Proprietary systems are "closed," requiring users to trust vendor blindly.
- **Support:** Git enjoys a massive global community for support while proprietary systems rely on official often paid helpdesks.
- **Freedom:** Git gives freedom for modifying and redistributing its software whereas proprietary software is strictly limited by legal contracts.

SHELL SCRIPTS DOCUMENTATION

To demonstrate practical application of concepts discussed in this audit about five shell scripts were developed to automate Linux administrative tasks.

- **Script 1 (System Identity):** Uses variables and command substitution to display real time system health and identity.

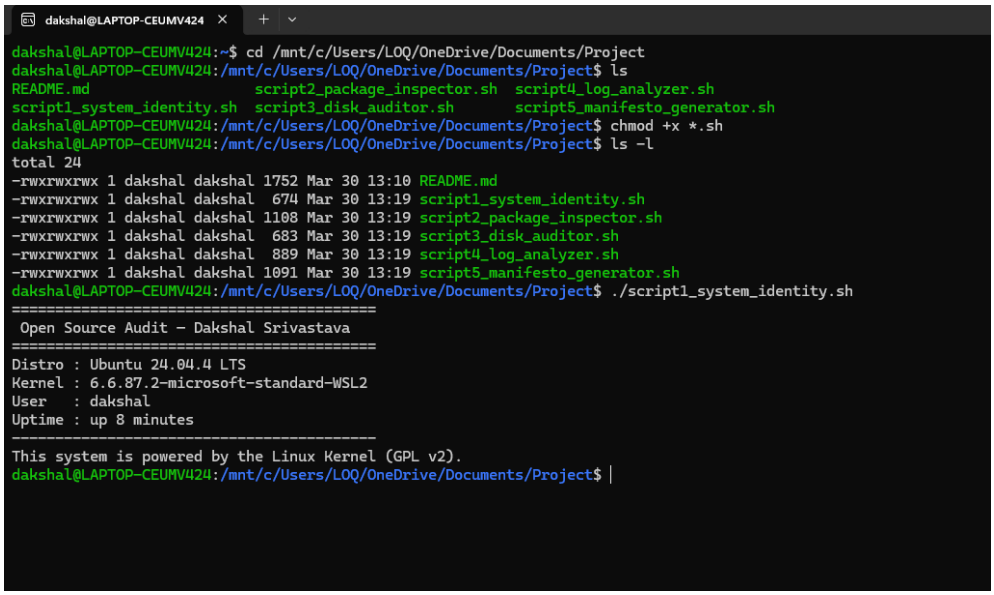


Figure 2: Execution of Script 1 (System Identity) verifying the Ubuntu 24.04 LTS environment and student credentials.

- **Script 2 (Package Inspector):** Implements conditional logic and case statements to verify the installation of Git and other critical packages.

```
dakshal@LAPTOP-CEUMV424
-rwxrwxrwx 1 dakshal dakshal 1752 Mar 30 13:10 README.md
-rwxrwxrwx 1 dakshal dakshal  674 Mar 30 13:19 script1_system_identity.sh
-rwxrwxrwx 1 dakshal dakshal 1108 Mar 30 13:19 script2_package_inspector.sh
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-rwxrwxrwx 1 dakshal dakshal 1091 Mar 30 13:19 script5_manifesto_generator.sh
dakshal@LAPTOP-CEUMV424:/mnt/c/Users/LOQ/OneDrive/Documents/Project$ ./script1_system_identity.sh
==========================================
 Open Source Audit - Dakshal Srivastava
==========================================
Distro : Ubuntu 24.04.4 LTS
Kernel : 6.6.87.2-microsoft-standard-WSL2
User   : dakshal
Uptime : up 8 minutes
------------------------------------------
This system is powered by the Linux Kernel (GPL v2).
dakshal@LAPTOP-CEUMV424:/mnt/c/Users/LOQ/OneDrive/Documents/Project$ ./script2_package_inspector.sh
[+] git is installed on this system.

--- Philosophy Note ---
Git: Linus Torvalds' tool that decentralized code collaboration and broke proprietary locks.
dakshal@LAPTOP-CEUMV424:/mnt/c/Users/LOQ/OneDrive/Documents/Project$ ./script3_disk_auditor.sh
Directory Audit Report
----------------------
/etc => drwxr-xr-x | 4.3M
/var/log => drwxrwxr-x | 66M
/home => drwxr-xr-x | 484K
/usr/bin => drwxr-xr-x | 95M
/tmp => drwxrwxrwt | 24K
dakshal@LAPTOP-CEUMV424:/mnt/c/Users/LOQ/OneDrive/Documents/Project$
```

Figure 3: Execution of Script 2 confirming Git installation and displaying the associated open-source philosophy note.

- **Script 3 (Disk Auditor):** Utilizes for loops and awk to scan system directories and extract human-readable permission and size data.

```
dakshal@LAPTOP-CEUMV424
dakshal@LAPTOP-CEUMV424:~$ cd /mnt/c/Users/LOQ/OneDrive/Documents/Project
dakshal@LAPTOP-CEUMV424:/mnt/c/Users/LOQ/OneDrive/Documents/Project$ ls
README.md                   script2_package_inspector.sh  script4_log_analyzer.sh
script1_system_identity.sh  script3_disk_auditor.sh       script5_manifesto_generator.sh
dakshal@LAPTOP-CEUMV424:/mnt/c/Users/LOQ/OneDrive/Documents/Project$ chmod +x *.sh
dakshal@LAPTOP-CEUMV424:/mnt/c/Users/LOQ/OneDrive/Documents/Project$ ls -l
total 24
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dakshal@LAPTOP-CEUMV424:/mnt/c/Users/LOQ/OneDrive/Documents/Project$ ./script1_system_identity.sh
==========================================
 Open Source Audit - Dakshal Srivastava
==========================================
Distro : Ubuntu 24.04.4 LTS
Kernel : 6.6.87.2-microsoft-standard-WSL2
User   : dakshal
Uptime : up 8 minutes
------------------------------------------
This system is powered by the Linux Kernel (GPL v2).
dakshal@LAPTOP-CEUMV424:/mnt/c/Users/LOQ/OneDrive/Documents/Project$ ./script2_package_inspector.sh
[+] git is installed on this system.

--- Philosophy Note ---
Git: Linus Torvalds' tool that decentralized code collaboration and broke proprietary locks.
dakshal@LAPTOP-CEUMV424:/mnt/c/Users/LOQ/OneDrive/Documents/Project$
```

Figure 4: Execution of Script 3 performing an automated permission and size audit of critical system directories.

- **Script 4 (Log Analyzer):** Reads system log files line by line using while read loops to identify and count error keywords.

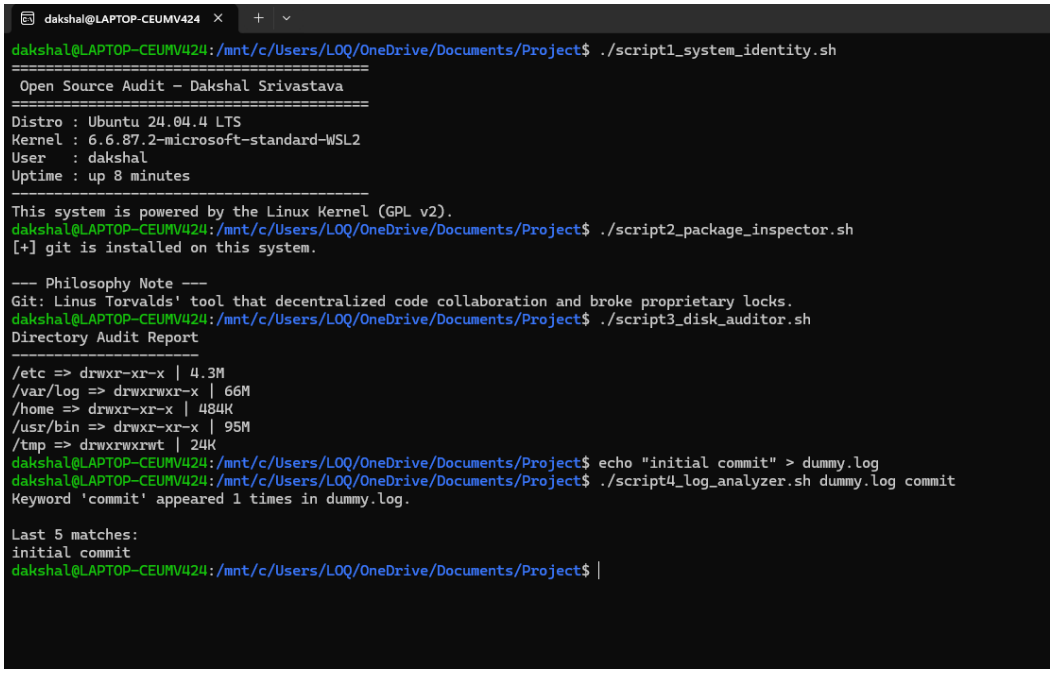

Figure 5: Execution of Script 4 demonstrating keyword searching and the automated 'tail' match extraction from a log file.

- **Script 5 (Manifesto Generator):** Takes user's input and generates a personalized statement of open source philosophy.

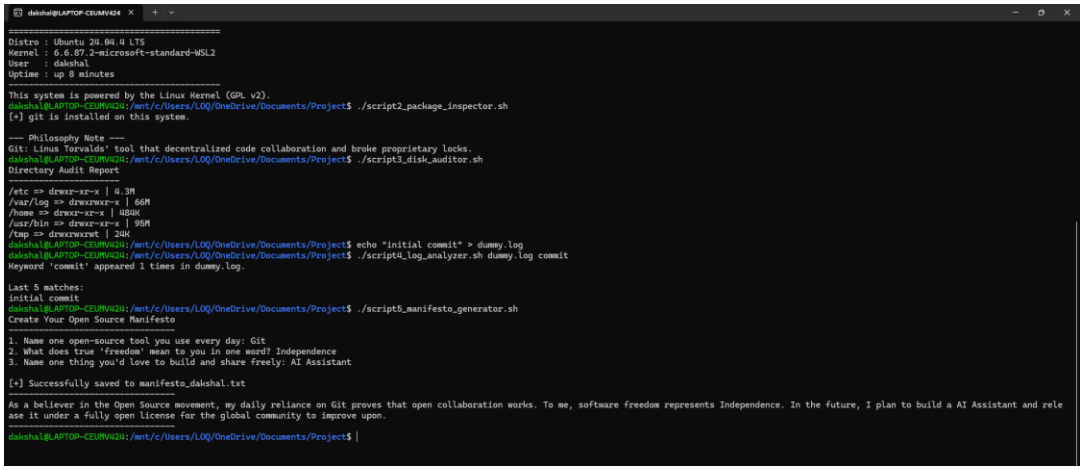

Figure 6: Interactive execution of Script 5 and the final generation of the personalized Open Source Manifesto.

CONCLUSION

In final analysis Git is far more than a simple version control utility; it is a foundational pillar of modern software engineering that embodies the very best principles of the open-source movement. Through the course of this audit, we have seen how Git was born from a specific crisis of proprietary restriction and grew into a global standard that empowers millions of developers through the protections of the GPL v2 license. It demonstrates with absolute clarity how open-source collaboration can solve real world technical problems such as the massive management requirements of the Linux kernel more effectively, securely and transparently than any closed, proprietary model could ever hope to achieve .

The lessons learned from studying Git extend far beyond the syntax of the command line they teach us about the power of transparency, the resilience of distributed systems, and the ethical importance of contributing back to the tools we use. Git proves that when code is shared freely, innovation accelerates because developers are no longer forced to work in silos or reinvent the wheel . Here they are part of a global conversation where every improvement benefits the collective whole.

As we move further into an era defined by cloud computing, artificial intelligence and decentralized software. Git remains an essential case study in how shared knowledge can accelerate human innovation. For any aspiring professional understanding Git is not just about learning how to manage code. It is about gaining a deeper insight into a collaborative spirit that values freedom over restriction and community over isolation. This project has underscored that the foundation of a technical education is not just the ability to write code but the ability to understand the history and philosophy of the tools that make that code

possible . As I move forward in my career, the principles of Git transparency, integrity and shared progress will remain the guiding framework for my professional development and contribution to the global software community.